Arithmetic sequences

1 By choosing any term after the first and subtracting the preceding term from it.

2

a No

b No

3

a $20, 24,$

b $-10, -14$

c $8, 9.5$

d $-\frac{20}{3}, -\frac{19}{3}$

4

a $d = -50$

b $d = \frac{3}{7}$

5

a

i $11, 14, 17, 20$

ii 3

b

i $11, 21, 31, 41$

ii 10

c

i $-7, -10, -13, -16$

ii -3

6

a $d = 0.9$

b $n = 11$

7 $-6, 0, 6, 12, 18$

8

a $n = 37$

b $n = 27$

c $n = 117$

d $n = 57$

9

a

i $d = 7$ ii $T_n = 7n - 2$ iii $T_{10} = 68$

b

i $d = -0.8$ ii $T_n = -0.8n + 17.8$

iii $T_10 = 9.8$

c

i $d = -7$ ii $T_n = -7n + 17$

iii $T_10 = -53$

d

i $d = \frac{3}{4}$ ii $T_n = \frac{3x + 17}{4}$

iii $T_10 = \frac{47}{4}$

Recursive rules

10 $T_n = T_{n-1} - 6, T_1 = 3$

11 $T_n = T_{n-1} + 25, T_1 = 30$

12

a $d = 6$ b $T_n = T_{n-1} + 6, T_1 = 2$

13

a $d = 5$ b $T_1 = 4$

c $T_n = T_{n-1} + 5, T_1 = 4$

Tables and graphs

14

a

n	1	2	3	4
T_n	1	5	9	13

b $d = 4$

15

a

n	1	3	5
T_n	-3	7	17

b $d = 5$

c $T_n = -8 + 5n$

d $T_{18} = 82$

16

a $d = 5$

b $T_n = -9 + 5n$

c $m = 5$

17

a

i $d = 6$

ii $T_n = 3 + 6n$

iii $T_{15} = 93$

b

i $d = -9$

ii $T_n = 14 - 9n$

iii $T_{10} = -76$

c

i $d = \frac{9}{2}$

ii $T_n = -\frac{1}{2} + \frac{9n}{2}$

iii $T_{11} = 49$

d

i $d = -8$

ii $T_n = -2 - 8n$

iii $T_{19} = -154$

e

i $d = -6$

ii $T_n = -4 - 6n$

iii $T_{18} = -112$

18

n	1	2	3	4	5
T_n	-6	-11	-16	-21	-26

19

a $d = 3$

b $T_n = -1 + 3n$

c $T_k = 1 - 3k$

Applications

20 $n = 32$

21

a $\$43\,920$

b $\frac{m}{2}(82\,080 + 1920m)$ dollars

c $y = 9$

22 $\$2670$

23

a

Day	left to pave (m^2)
1	800
2	750
3	700
4	650
5	600

b Linear decay

24

a 5800 phones

b $165\,600$ phones

25 1510 in

26

a $d = 35$

b $a = 151$

c $n = 22$

27

a $3 + n$

b $\frac{n}{2}(7 + n)$ pipes

28

a $6k + 2$ yd

b $n(3n + 5)$ yd

c $n = 10$

29

30

a 4.5 km

b 112.25 km

31

a 430 lb

b 5390 lb

c 26

32

a \$300

b \$2200

c \$1900

d $V_{n+1} = V_n + 300, V_0 = 1900$

33

a \$20

b \$1200

c $V_{n+1} = V_n - 20, V_0 = 1200$

34

a 50 m

b 200 m

c 500 m

35

a \$1972

b \$1860

c $T_n = T_{n-1} - 4, T_0 = 2000$

36

a \$21

b $T_n = T_{n-1} + 1.95, T_0 = 1.5$

37

a \$24.09

b $V_{n+1} = V_n - 2.05, V_0 = 42.54$

38

a \$365

b $u_n = u_{n-1} + 45, u_0 = 230$

c $v_n = v_{n-1} + 45, v_0 = 460$

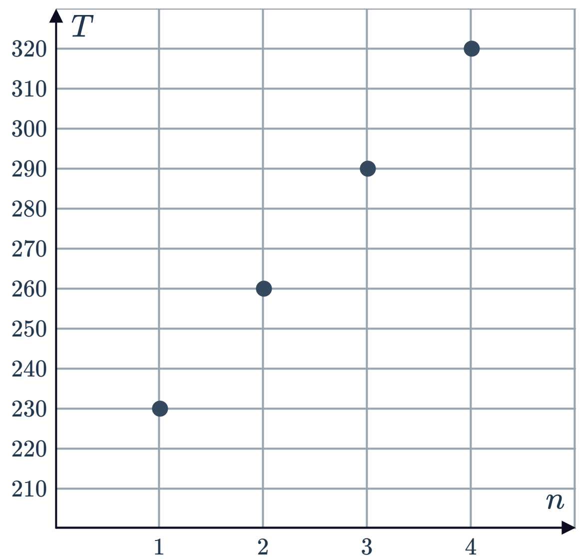
39

a

n	1	2	3	4	18
T	230	260	290	320	740

b 30

c



d A straight line